

INDIAN AND CHINESE DEVELOPMENT IN COMPARATIVE PERSPECTIVE

DATA ASSIGNMENT 2

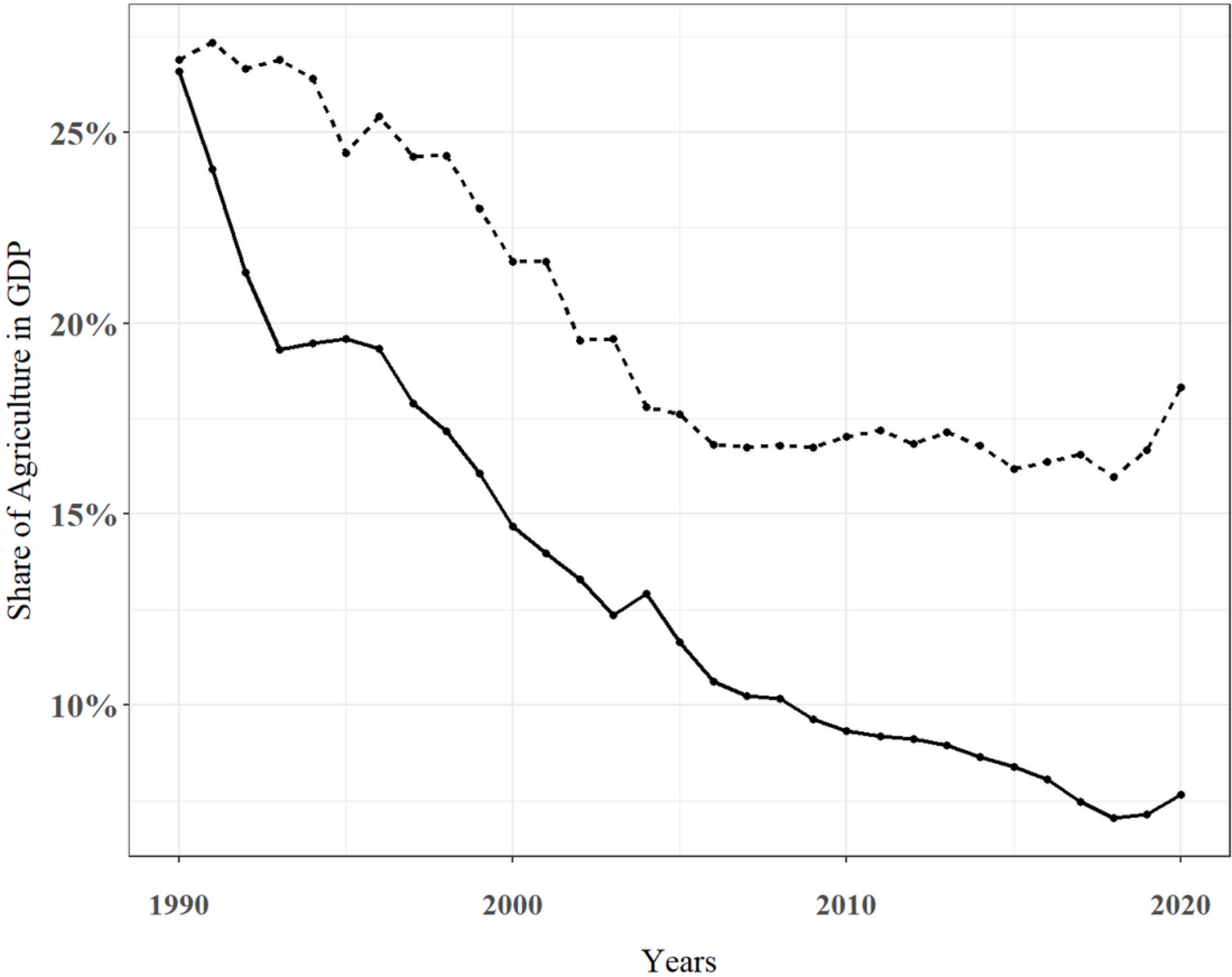
Vaibhav Agarwal

Agricultural Sector

**COMPARING CHANGES IN
AGRICULTURAL SHARE OF EMPLOYMENT
AND GDP FROM 1990 TO 2020 IN
INDIA AND CHINA**

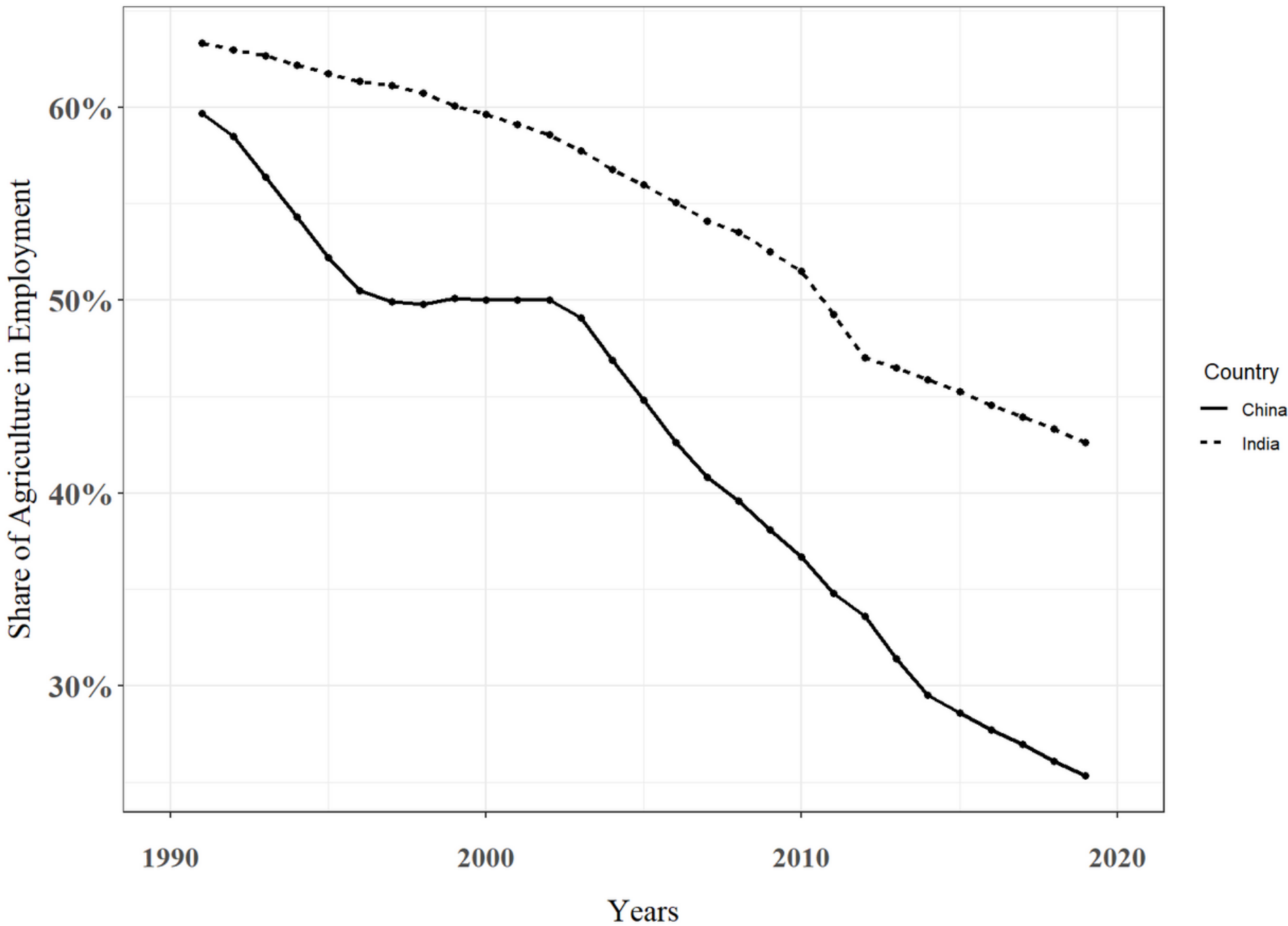
AGRICULTURAL SECTOR

Agriculture Share of GDP from 1990 to 2020



(I) AGRICULTURAL SHARE OF GDP

Agriculture Share of Employment from 1990 to 2020

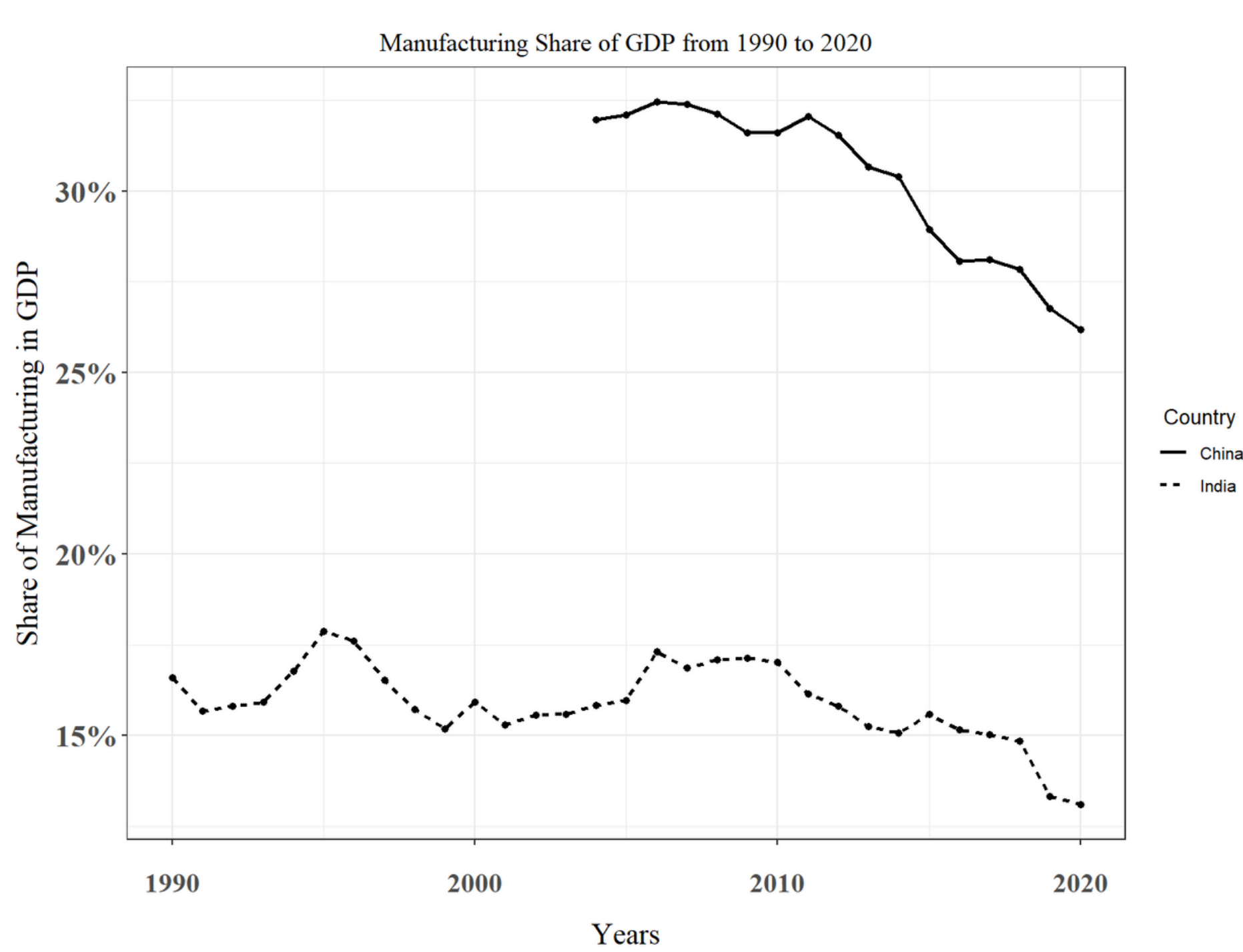


(II) AGRICULTURAL SHARE OF EMPLOYMENT

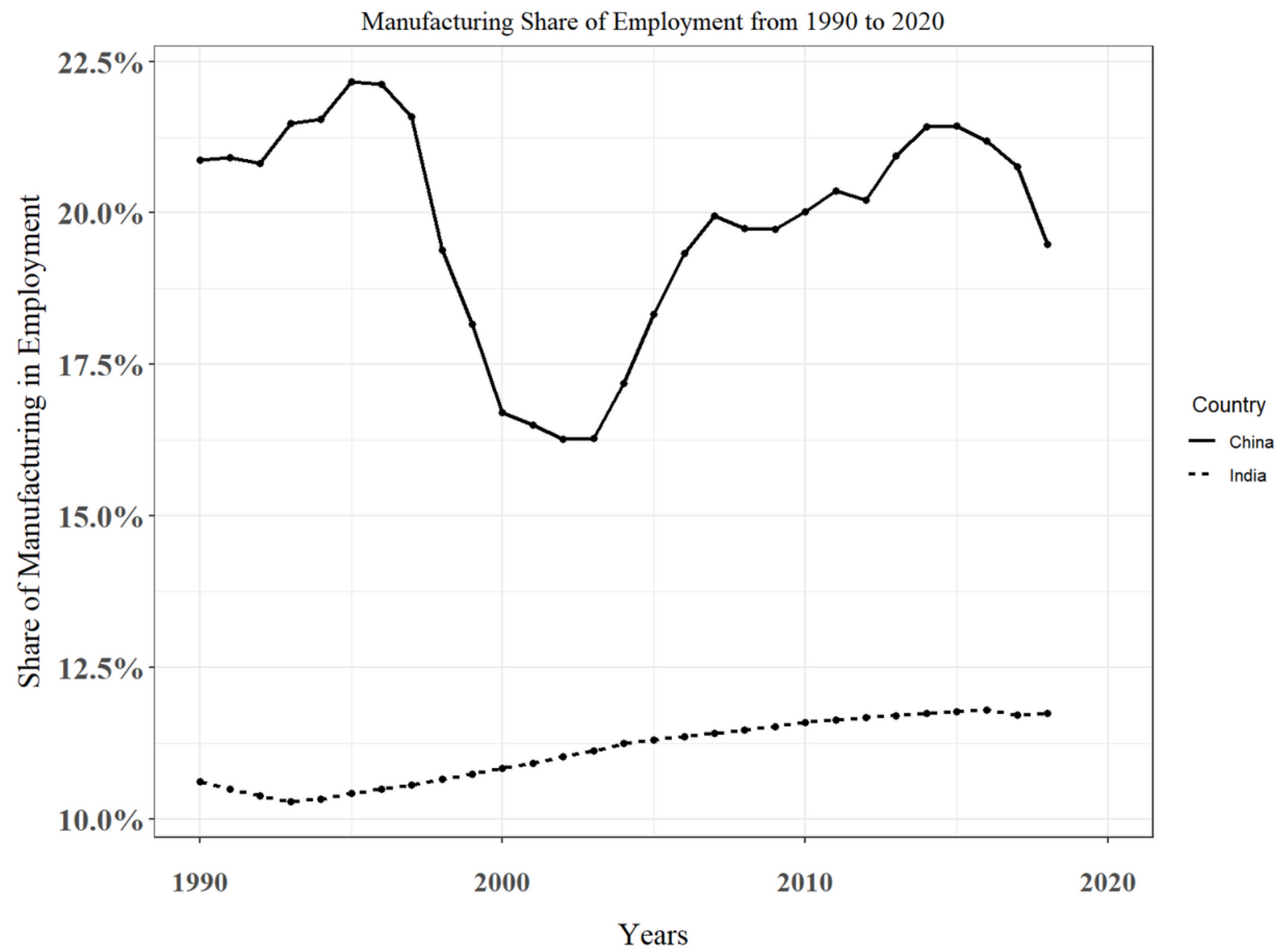
Manufacturing Sector

**COMPARING CHANGES IN
MANUFACTURING SHARE OF EMPLOYMENT
AND GDP FROM 1990 TO 2020 IN
INDIA AND CHINA**

MANUFACTURING SECTOR



(I) MANUFACTURING SHARE OF GDP

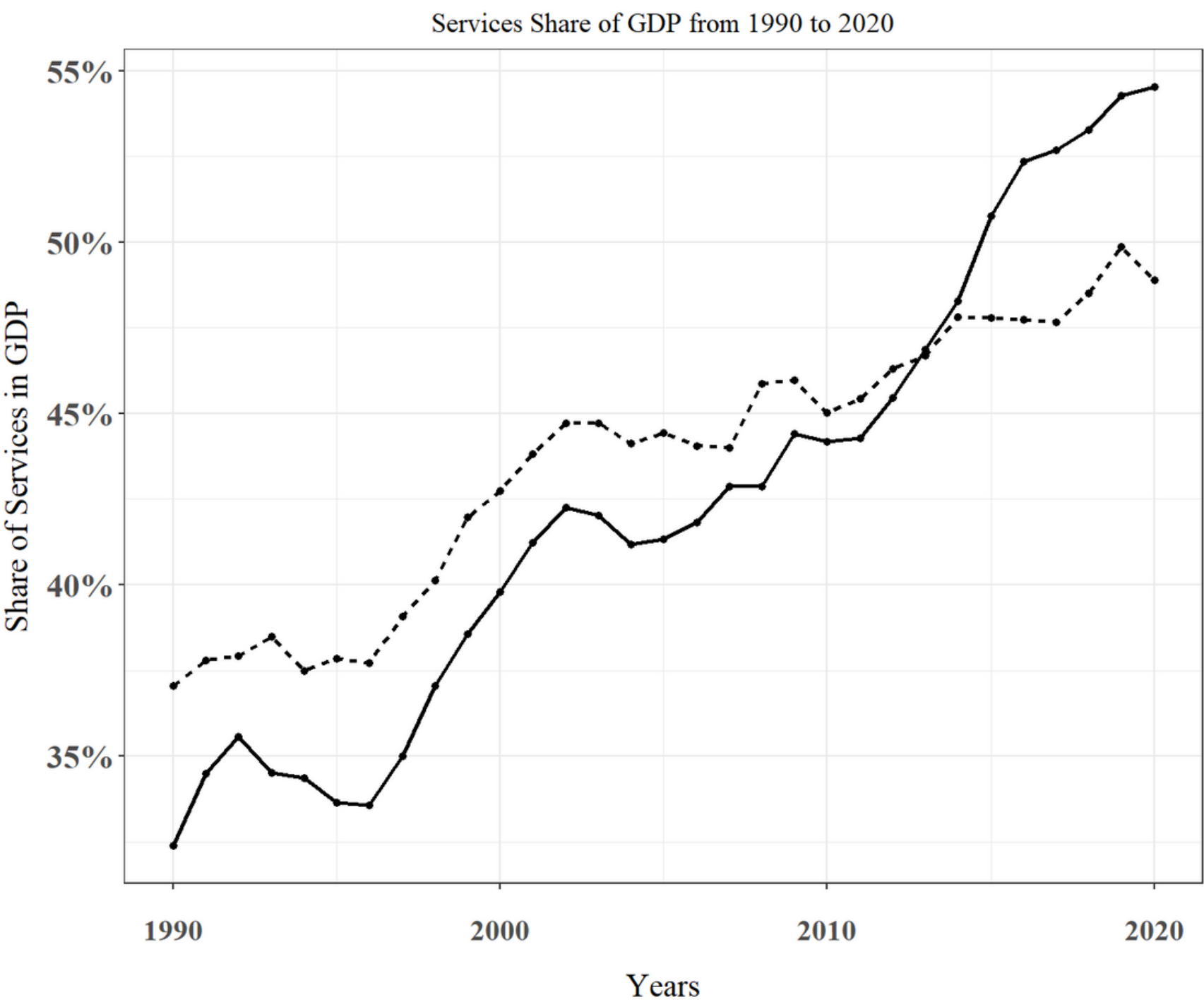


(II) MANUFACTURING SHARE OF EMPLOYMENT

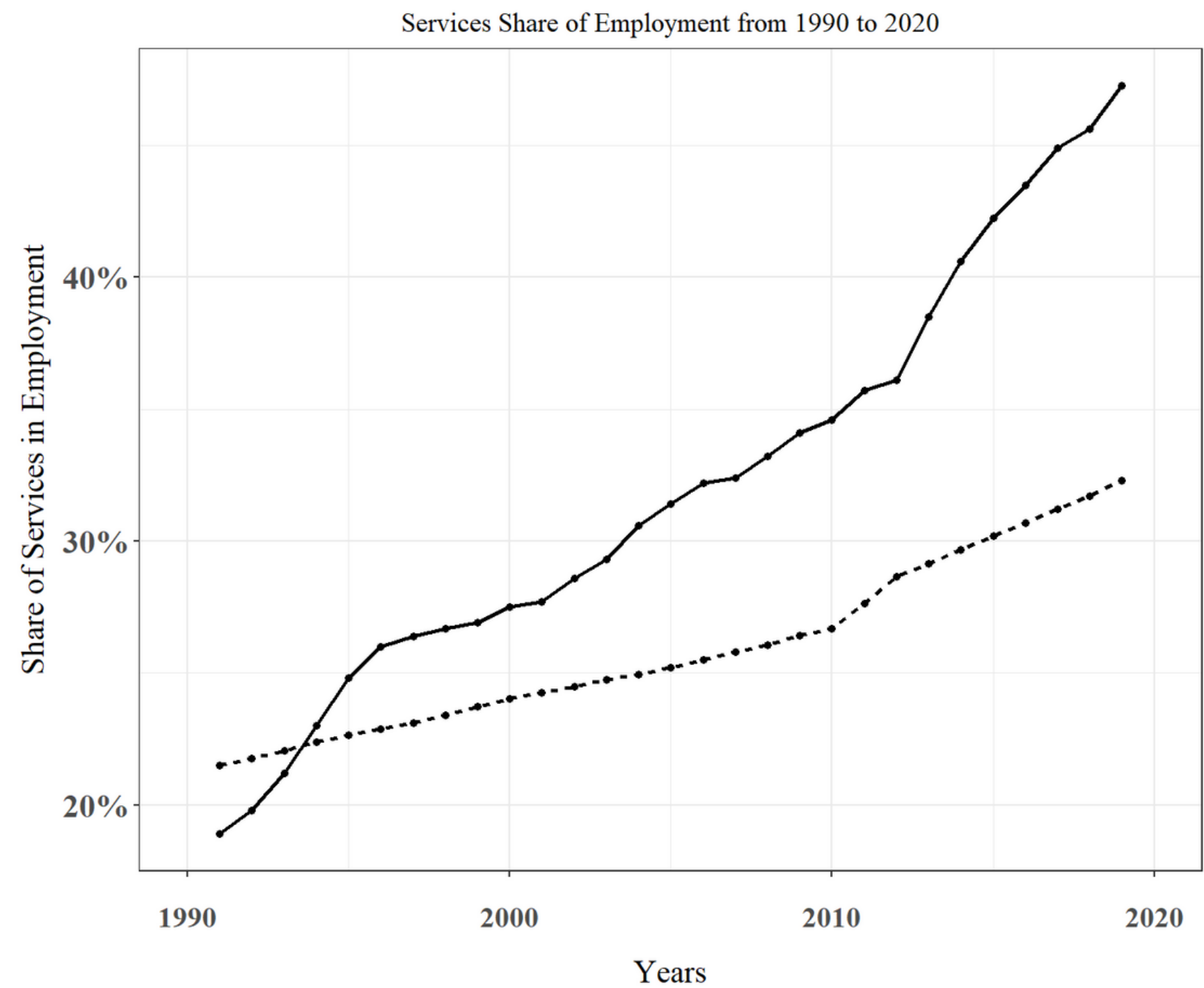


**COMPARING CHANGES IN SERVICES
SHARE OF EMPLOYMENT AND GDP FROM
1990 TO 2020 IN
INDIA AND CHINA**

SERVICES SECTOR



(I) SERVICES SHARE OF GDP



(II) SERVICES SHARE OF EMPLOYMENT

At a Glance

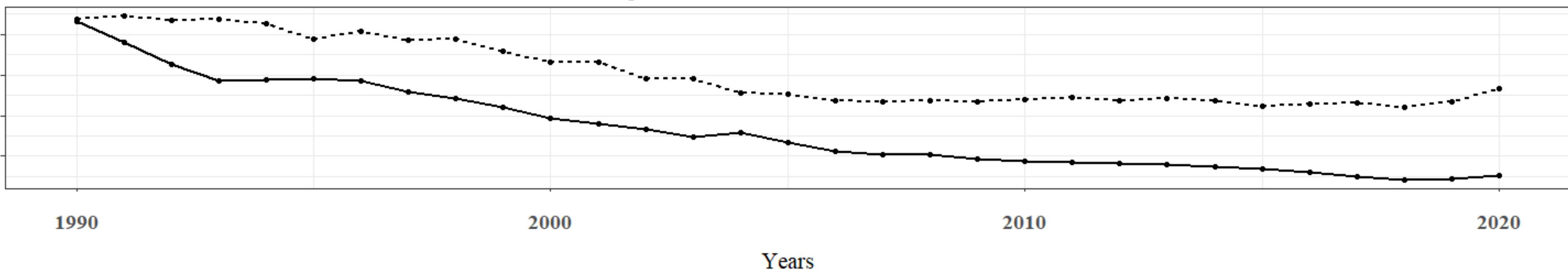
COMPARING CHANGES IN **SECTORAL
SHARES OF GDP FROM 1990 TO 2020 IN
INDIA AND **CHINA****

Share of Agriculture in GDP

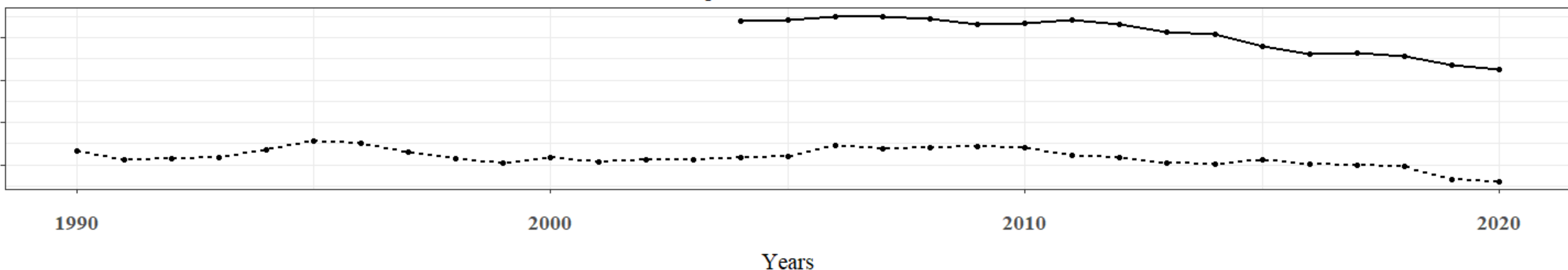
Share of Manufacturing in GDP

Share of Services in GDP

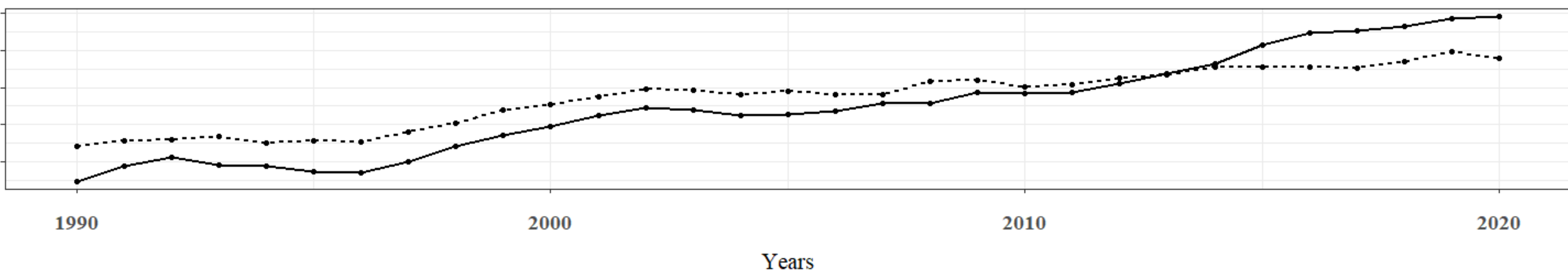
Agriculture Share of GDP from 1990 to 2020



Manufacturing Share of GDP from 1990 to 2020



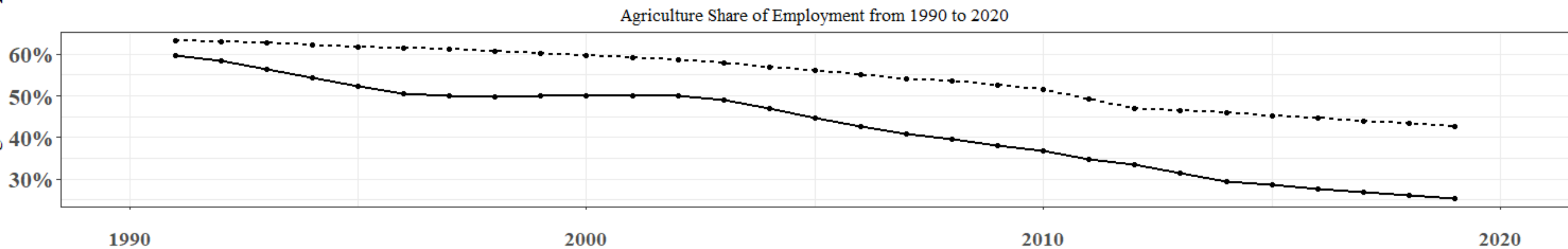
Services Share of GDP from 1990 to 2020



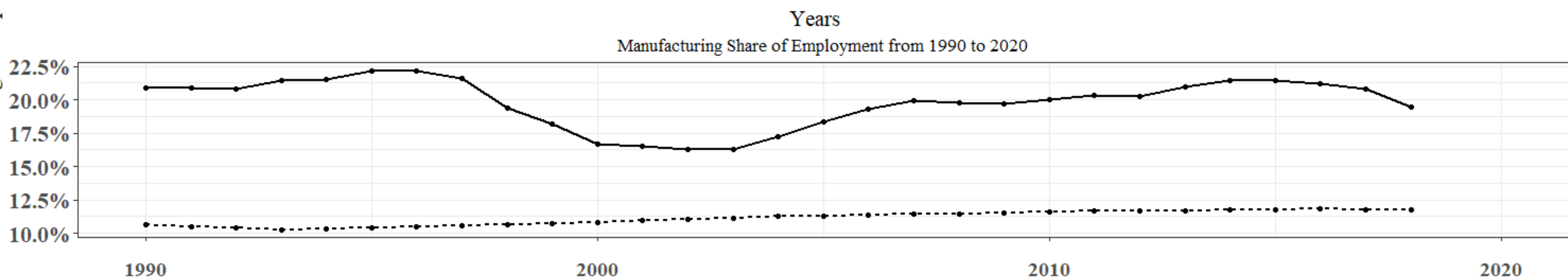
At a Glance

COMPARING CHANGES IN **SECTORAL
SHARES OF EMPLOYMENT FROM 1990 TO
2020 IN
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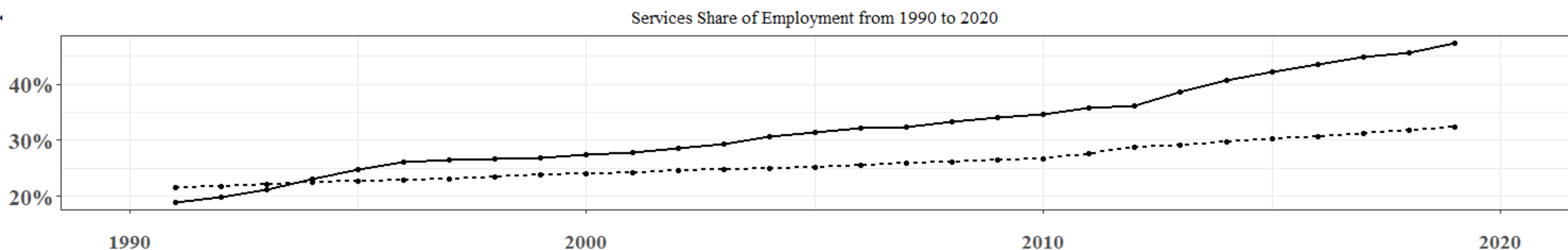
Share of Agriculture in Emp



Share of Manufacturing in Emp



Share of Services in Emp

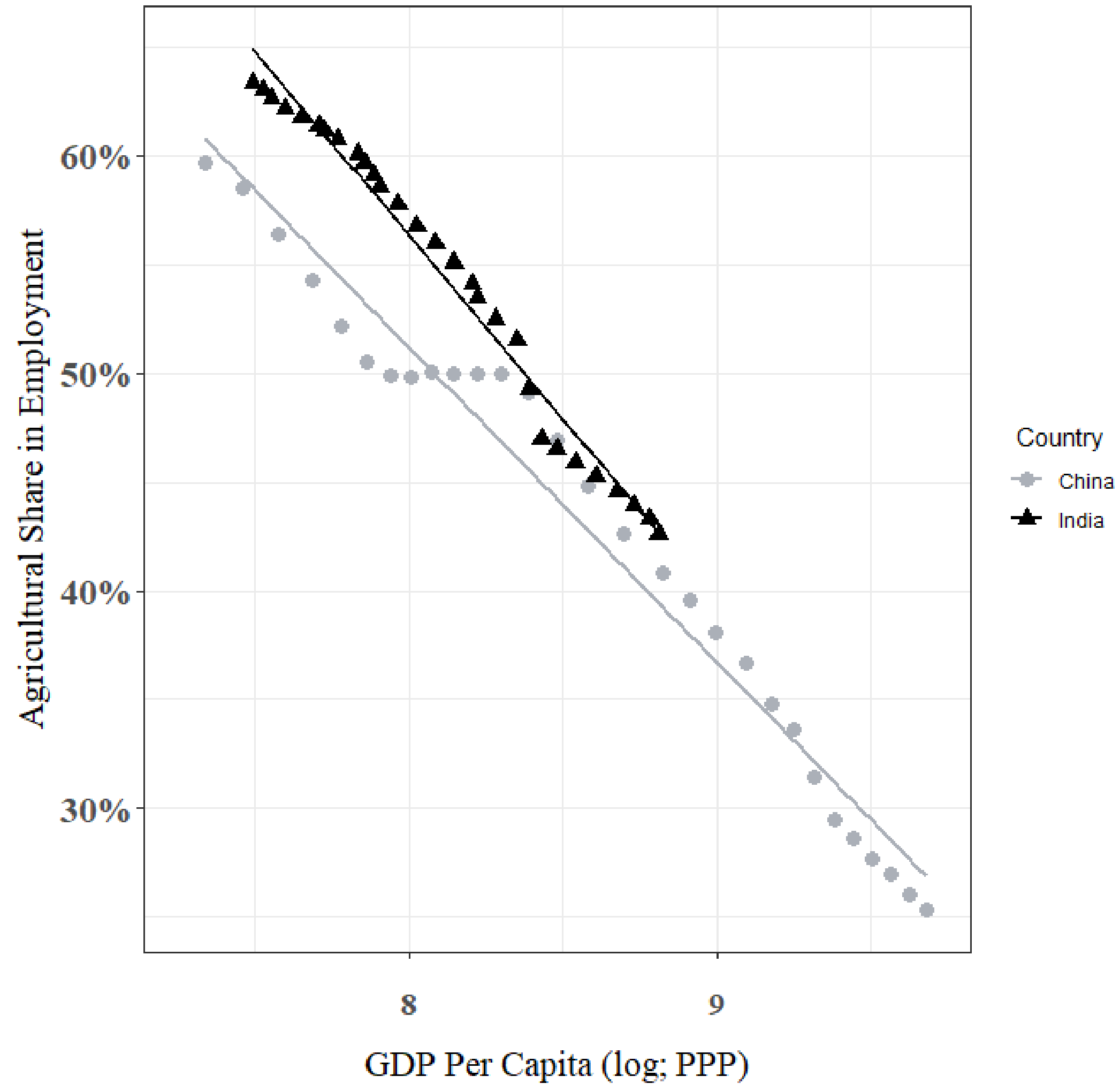


Years

Agriculture & GDP

**GRAPHICALLY REPRESENTING THE
AGRICULTURAL SHARE OF EMPLOYMENT
AS A FUNCTION OF GDP PER CAPITA
(PPP) FOR
INDIA AND CHINA**

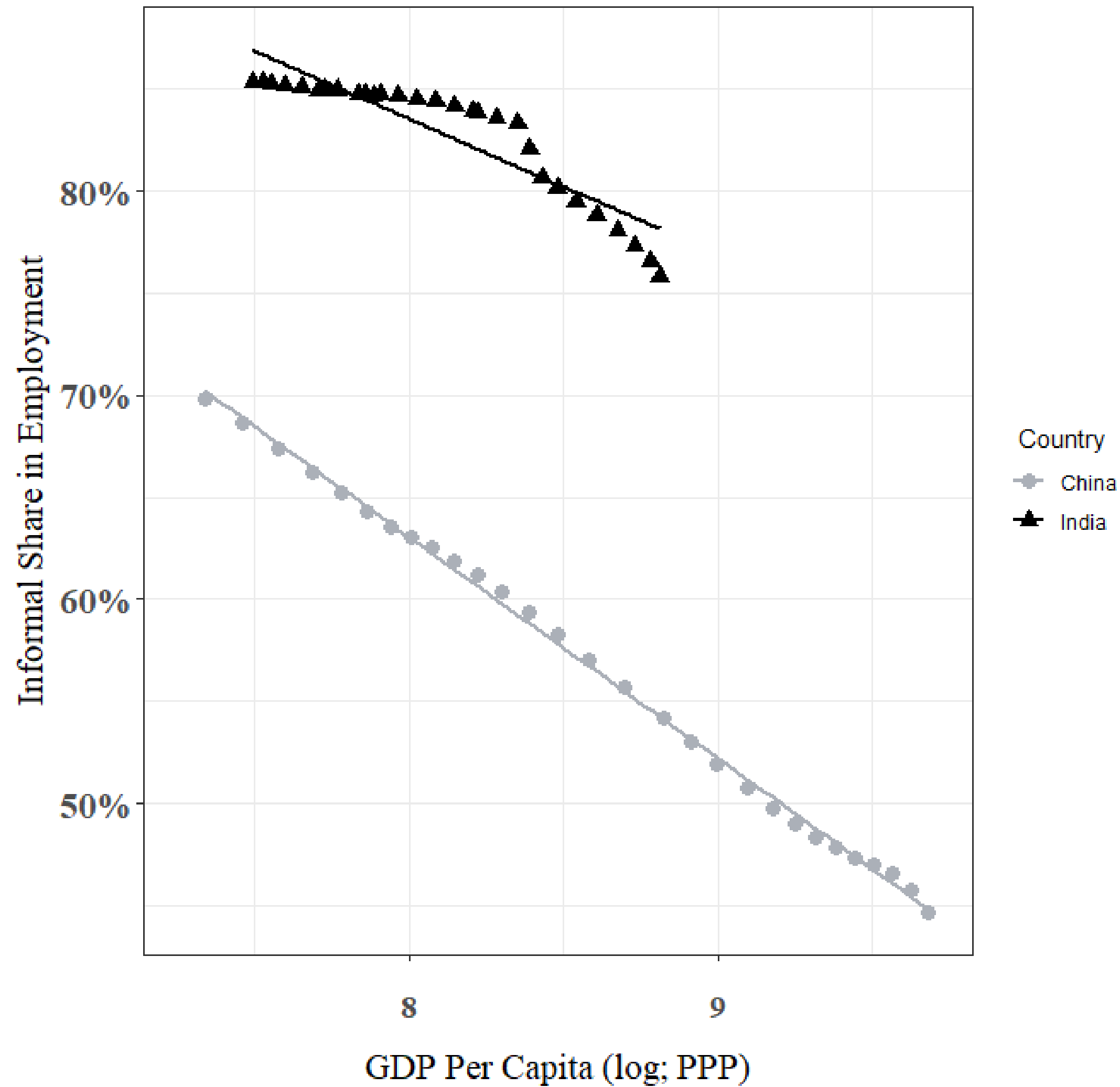
Looking at the Relation Between Agricultural Share in Employment and GDP Per Cap



Informal Share & GDP

**GRAPHICALLY REPRESENTING THE
INFORMAL SHARE OF EMPLOYMENT AS A
FUNCTION OF GDP PER CAPITA (PPP)
FOR INDIA AND CHINA**

Looking at the Relation Between Informal Share in Employment and GDP Per Cap



Regression

REGRESSING
INFORMAL SHARE OF EMPLOYMENT
AGAINST
GDP PER CAPITA (PPP)
FOR INDIA AND CHINA

Table 1: Regressing Informal share of employment on
GDP Per Capita (log; PPP)

	<i>Dependent variable:</i>	
	Informal (China)	Informal (India)
	(1)	(2)
China GDP (log PC; PPP)	−0.108*** (0.001)	
India GDP (log PC; PPP)		−0.067*** (0.006)
Constant	1.497*** (0.008)	1.368*** (0.047)
Observations	29	29
R ²	0.998	0.831
Adjusted R ²	0.998	0.825
Residual Std. Error (df = 27)	0.004	0.013
F Statistic (df = 1; 27)	12,909.550***	132.654***

Note:

*p<0.1; **p<0.05; ***p<0.01



INTERPRETATION

A simple linear-log OLS Regression model of with **Informal Share of Employment regressed against GDP Per capita in PPP log transformed** is used. The coefficient attached with GDP per capita is our coefficient of interest; it is **statistically significant at the 99% confidence interval** with a p-value less than 0.01. The **coefficient is negative** for both nations; indicating that **a increase in GDP per capita causes the informal share of employment to fall**.

A 100% rise (or doubling) of the GDP per capita will result in a fall in informal share of employment by 7.49% $((\log(2) * 0.108) * 100)$ in China, whereas, for India, a 4.64% $((\log(2) * 0.067) * 100)$ fall would be observed. (Since the value of informal share of employment lie between 0 and 1, we must multiply the coefficient values by 100 to be able to interpret the coefficients.)

We can say that China has been able to **reduce it's informal share of employment at a rate faster than that of India as GDP per capita rises**.

Regression

REGRESSING
AGRICULTURAL SHARE OF EMPLOYMENT
AGAINST
GDP PER CAPITA (PPP)
FOR INDIA AND CHINA

Table 2: Regressing Agricultural share of employment on
GDP Per Capita (PPP)

	<i>Dependent variable:</i>	
	Agri (China)	Agri (India)
	(1)	(2)
China GDP (log PC; PPP)	−0.145*** (0.005)	
India GDP (log PC; PPP)		−0.169*** (0.005)
Constant	1.669*** (0.041)	1.915*** (0.039)
Observations	29	29
R ²	0.972	0.979
Adjusted R ²	0.971	0.978
Residual Std. Error (df = 27)	0.018	0.010
F Statistic (df = 1; 27)	928.094***	1,257.596***

Note:

*p<0.1; **p<0.05; ***p<0.01



INTERPRETATION

A simple linear-log OLS Regression model of with **Agricultural Share of Employment regressed against GDP Per capita in PPP log transformed** is used. The coefficient attached with GDP per capita is our coefficient of interest; it is **statistically significant at the 99% confidence interval** with a p-value less than 0.01. The **coefficient is negative** for both nations; indicating that **a increase in GDP per capita causes the agricultural share of employment to fall**.

A 100% rise (or doubling) of the GDP per capita will result in a fall in agricultural share of employment by 10% $((\log(2) * 0.145) * 100)$ in China, whereas, for India, a 11.71% $((\log(2) * 0.169) * 100)$ fall in agricultural share would be observed. (Since the value of agricultural share of employment lie between 0 and 1, we must multiply the coefficient values by 100 to be able to interpret the coefficients.)

We can say that India has been able to **reduce it's agricultural share of employment at a rate faster than that of China as GDP per capita rises**.

STRUCTURAL TRANSFORMATION IN INDIA AND CHINA



STRUCTURAL TRANSFORMATION IN INDIA AND CHINA IN A COMPARITIVE PERSPECTIVE

India and China have both undergone structural transformation over the last couple of decades; the agriculture share of employment has declined in both countries with modest increases in manufacturing and a shift from agriculture to services.

In the case of China, the manufacturing share of employment and GDP has declined which is in line with the present consensus on a Kuznets shaped curve, stating that the share of employment in manufacturing tends to increase as an economy develops, reaches a peak, and then decline as the economy becomes more service-oriented. Structural transformation has taken place in China with clear shifts towards deindustrialization and a services oriented economy; however, having undergone a shift from agriculture to manufacturing prior to that. The Chinese economy has also been able to reduce its share of informal share of employment as GDP per capita rise at a rate greater than that of India.



STRUCTURAL TRANSFORMATION IN INDIA AND CHINA IN A COMPARITIVE PERSPECTIVE

India on the other hand, has undergone, as Dani Rodrik points out "premature deindustrialization". Rodrik points out that many developing countries are experiencing a decline in manufacturing employment and output at an earlier stage of development than was seen in developed countries, hence the term premature deindustrialization (the peak share of manufacturing has fallen; it has come earlier as well). He suggested that this premature deindustrialization may be a result of globalization and economies of scale, where, developing countries can no longer manufacture goods at a prices which can compete foreign goods. Despite having undergone a shift from the agricultural sector faster than that of China, India has skipped the phase of rise in manufacturing industries and is now a witnessing a decline in manufacturing or prematurely deindustrialization. This points towards concerns regarding the development of infrastructure owing to backward linkages of manufacturing industries as well as possible slowdown in economic growth arising from a deviation from it's compactivitie advantage.